

ALEXANDER CITY, AL, USA

WMO: 720265

Lat: 32.915N

Lon: 85.963W

Elev: 209

StdP: 98.84

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63833

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.7	(c) -2.3	(d) -15.1	(e) 1.0	(f) 4.3	(g) -11.9	(h) 1.4	(i) 2.1	(j) 8.7	(k) 7.1	(l) 7.9	(m) 7.2	(n) 1.8	(o) 350	(p) 0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	35.1	24.2	33.9	24.3	32.8	24.1	27.2	31.6	26.5	30.9	25.7	30.2	2.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.0	29.6	25.2	20.9	28.6	24.7	20.2	28.0	86.9	31.7	84.0	30.8	80.4	30.3	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.2	(b) 5.8	(c) 5.0	(e) -8.5	(f) 35.4	(g) 2.3	(h) 1.3	(i) -10.1	(j) 36.4	(k) -11.5	(l) 37.1	(m) -12.8	(n) 37.9	(o) -14.4	(p) 38.9		
(a) 7.2	(b) 5.8	(c) 5.0	(e) -9.4	(f) 27.9	(g) 2.2	(h) 1.1	(i) -11.0	(j) 28.7	(k) -12.3	(l) 29.4	(m) -13.5	(n) 30.0	(o) -15.2	(p) 30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	7.6	10.0	13.6	17.5	21.8	26.0	27.1	26.8	24.5	18.4	12.3	9.5		
	DBStd	7.95	5.48	4.98	4.68	3.64	3.26	2.09	1.67	1.87	2.91	4.40	4.35	4.97		
	HDD10.0	291	113	57	21	1	0	0	0	0	0	2	24	72		
	HDD18.3	1311	333	235	157	58	10	0	0	0	2	55	186	275		
	CDD10.0	3198	39	58	131	226	364	479	531	521	435	261	94	58		
	CDD18.3	1176	1	3	9	33	116	229	273	262	187	56	6	2		
	CDH23.3	10622	3	8	95	332	1031	2177	2604	2381	1532	418	34	8		
	CDH26.7	4375	0	0	11	51	353	998	1183	1058	620	99	3	0		
(14) Wind	WSAvg	1.6	2.1	2.1	2.2	1.9	1.5	1.2	1.2	1.1	1.3	1.5	1.6	1.8		
(15) Precipitation	PrecAvg	1305	132	128	145	112	91	115	134	100	93	64	113	109		
	PrecMax	1579	230	224	320	207	250	243	262	168	200	229	281	178		
	PrecMin	754	61	57	33	58	6	47	43	51	21	6	57	71		
	PrecStd	190	49	47	70	41	55	62	57	37	48	50	55	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	24.8	27.9	30.6	33.0	36.4	36.1	35.6	35.1	32.1	26.5	24.4		
		MCWB	17.4	19.0	16.5	19.3	22.2	23.3	24.8	24.1	24.3	23.2	19.7	20.2		
	2%	DB	20.9	22.6	26.1	27.9	32.1	34.6	34.7	34.9	33.2	29.1	24.0	22.0		
		MCWB	16.8	17.0	16.7	18.1	21.9	23.7	24.7	24.5	23.5	21.4	17.2	18.1		
	5%	DB	19.7	20.7	23.9	27.0	30.9	33.0	33.3	33.0	32.3	27.3	22.5	20.2		
		MCWB	15.9	15.6	15.9	17.8	21.7	24.0	24.6	24.6	24.0	19.9	17.3	17.7		
	10%	DB	16.6	19.0	22.2	25.4	28.4	32.1	32.5	32.3	31.0	25.9	20.8	18.2		
		MCWB	14.0	15.1	15.1	17.4	20.6	23.6	24.6	24.8	23.2	19.2	15.9	15.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	20.4	20.2	22.6	24.7	27.5	27.9	28.4	26.9	24.5	22.5	21.4		
		MCDB	21.1	23.2	22.3	26.7	29.0	31.7	32.1	32.5	31.5	28.2	23.6	23.1		
	2%	WB	18.1	19.0	19.2	21.2	23.6	26.4	27.0	27.3	25.9	23.5	20.5	19.8		
		MCDB	20.2	21.1	22.2	25.0	28.7	30.5	31.5	31.6	30.3	26.8	22.5	20.9		
	5%	WB	16.5	17.7	18.2	20.1	22.9	25.4	26.2	26.5	25.1	22.5	19.0	18.6		
		MCDB	18.3	19.7	21.7	23.8	27.7	30.4	30.7	30.9	29.2	25.4	21.4	20.0		
	10%	WB	14.6	15.6	17.0	19.3	22.2	24.7	25.4	25.7	24.1	21.4	17.1	16.0		
		MCDB	16.5	17.6	21.3	23.1	26.6	29.8	30.2	29.9	28.0	24.0	19.4	17.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	11.9	12.7	13.3	12.1	11.0	10.2	9.8	10.7	12.4	13.2	10.9		
		MCDBR	13.0	13.6	14.7	14.5	14.4	12.5	11.5	11.2	12.8	13.7	14.4	11.7		
	5% WB	MCWBR	9.2	8.6	7.3	6.8	6.1	5.2	4.5	4.7	5.1	6.7	8.8	8.9		
		MCWBR	10.3	10.6	11.1	11.1	11.1	10.8	10.2	9.9	10.3	9.9	10.5	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.323	0.354	0.386	0.446	0.477	0.524	0.521	0.423	0.368	0.341	0.322		
	taud		2.527	2.512	2.439	2.337	2.211	2.175	2.054	2.052	2.317	2.458	2.472	2.529		
	Ebn at Noon		903	934	924	903	848	818	777	773	842	871	865	877		
	Edh at Noon		84	94	109	126	144	149	168	166	122	98	88	80		
(44) All-Sky Solar Radiation	RadAvg		2.73	3.28	4.42	5.58	6.13	6.14	5.97	5.50	4.84	4.10	3.08	2.33		
	RadStd		0.18	0.41	0.39	0.39	0.48	0.47	0.41	0.32	0.43	0.52	0.30	0.25		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+104

Nomenclature: See separate page